UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES

Software Requirements Specification(SRS)

For

All In One Dictionary

A Multilingual Smart Dictionary System

Version: 1.0 Approved

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Revision History

Version	Date	Description	Author
1.0	2026	SRS Creation	Project Team

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of All In One Dictionary, a multilingual smart dictionary system.

The system allows users to search words and view meanings, pronunciation, translations, and example usage across multiple languages within a single application.

1.2 Document Conventions

- ➢ REQ-XX → Functional Requirement
- ➢ SHALL → Mandatory requirement
- ➢ MAY → Optional feature
- ➢ Priority Levels:
  - ○ High
  - ○ Medium
  - ○ Low

1.3 Intended Audience

❒This document is intended for:

- Developers — system implementation
- Instructors — project evaluation
- Testers — system validation
- Users — system understanding
- Documentation writers

❒Reading Order:
Introduction → Overall Description → System Features → Non-Functional Requirements.

1.4 Product Scope

❒All In One Dictionary provides:

- Multilingual word meanings
- Pronunciation support
- Example sentences
- Search history tracking
- Favorite word saving
- Offline dictionary access

➢Goal:
Create a unified educational dictionary platform eliminating the need for multiple language apps.

1.5 References

- ❖ IEEE SRS Template — Karl E. Wiegers
- ❖ Software Engineering — Ian Sommerville
- ❖ HTML/CSS/JavaScript Documentation
- ❖ MySQL Documentation
- ❖ Node.js Documentation

2. Overall Description

2.1 Product Perspective

The system is a **web-based dictionary application**.

❒System Architecture:

User → Web Interface → Dictionary Engine → Database/API → Result Output

❒The application integrates:

- Local dictionary database
- Translation API
- Text-to-Speech API

2.2 Product Functions

❒Major functions include:

- Word search
- Multilingual translation
- Pronunciation playback
- Example sentence display

- Favorite word storage
- Search history management
- Offline access

2.3 User Classes and Characteristics

1. Students (Primary Users)

- Frequent use
- Language learning purpose

2. General Users

- Casual word searching

3. Administrator

- Maintains dictionary database
- Updates language data

2.4 Operating Environment

- Platform: Web Browser / Mobile Browser
- OS: Windows, Linux, Android
- Browser: Chrome, Edge, Firefox
- Internet required for online translation

2.5 Design and Implementation Constraints

- Must use HTML, CSS, JavaScript
- SQL database required
- API dependency exists
- Unicode language support required

2.6 User Documentation

➢Delivered documentation:

- User Manual
- Installation Guide
- Online Help System
- Tutorial Guide

2.7 Assumptions and Dependencies

➢Assumptions:

- Users have basic internet access.
- APIs remain available.

➢Dependencies:

- Translation API
- Text-to-Speech API
- Database server

3. External Interface Requirements

3.1 User Interfaces

❒Interface Components:

- Search Bar
- Language Selector
- Meaning Display Panel
- Audio Pronunciation Button
- Favorites Panel
- History Panel

❒UI Requirements:

- Responsive design

- Clean layout
- Dark mode support

3.2 Hardware Interfaces

❒Supported Devices:

- Desktop
- Laptop
- Smartphone
- Tablet

➢Input Methods:

- Keyboard
- Mouse
- Touchscreen

3.3 Software Interfaces

❒Interfaces with:

- MySQL Database
- Translation API
- Text-to-Speech Service
- Web Browser Environment

3.4 Communications Interfaces

❒Protocols Used:

- HTTP/HTTPS
- REST API communication
- Secure encrypted data transfer

4. System Features

4.1 Smart Search Engine

❒Description

Allows users to search words instantly.

➢Priority: **High**

❒Stimulus/Response

User enters word → System processes → Meaning displayed.

❒Functional Requirements

- REQ-01: System shall accept word input.
- REQ-02: System shall validate input.
- REQ-03: System shall display meanings.
- REQ-04: System shall show error message for invalid search.

4.2 Multilingual Translation

➢Priority: **High**

- REQ-05: System shall display meanings in multiple languages.
- REQ-06: User shall select language.
- REQ-07: System shall fetch translations via API/database.

4.3 Pronunciation Tool

➢Priority: **Medium**

- REQ-08: System shall provide pronunciation audio.
- REQ-09: User shall play audio pronunciation.
- REQ-10: System shall use Text-to-Speech technology.

4.4 Example Sentence Feature

➢Priority: **Medium**

- REQ-11: System shall display example sentences.
- REQ-12: System shall support multilingual examples.

4.5 Favorites & History (Word Vault)

➢Priority: **Medium**

- REQ-13: User shall save favorite words.
- REQ-14: System shall maintain search history.
- REQ-15: User shall remove saved items.

4.6 Offline Mode

➢Priority: **Low**

- REQ-16: System shall allow offline access to stored words.
- REQ-17: Frequently searched words shall be cached locally.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Search result response ≤ 2 seconds.
- System shall handle multiple users simultaneously.
- Optimized database queries required.

5.2 Safety Requirements

- Prevent application crash.
- Regular database backup.
- Graceful handling of API failure.

5.3 Security Requirements

- Secure database storage.
- Encrypted communication.
- Admin authentication required.

5.4 Software Quality Attributes

- Usability — Easy interface
- Reliability — Stable operation
- Maintainability — Modular design
- Scalability — Add new languages easily
- Portability — Works across devices

5.5 Business Rules

- Only admin can update dictionary data.
- Unlimited search allowed.
- Educational usage focus.

6. Other Requirements

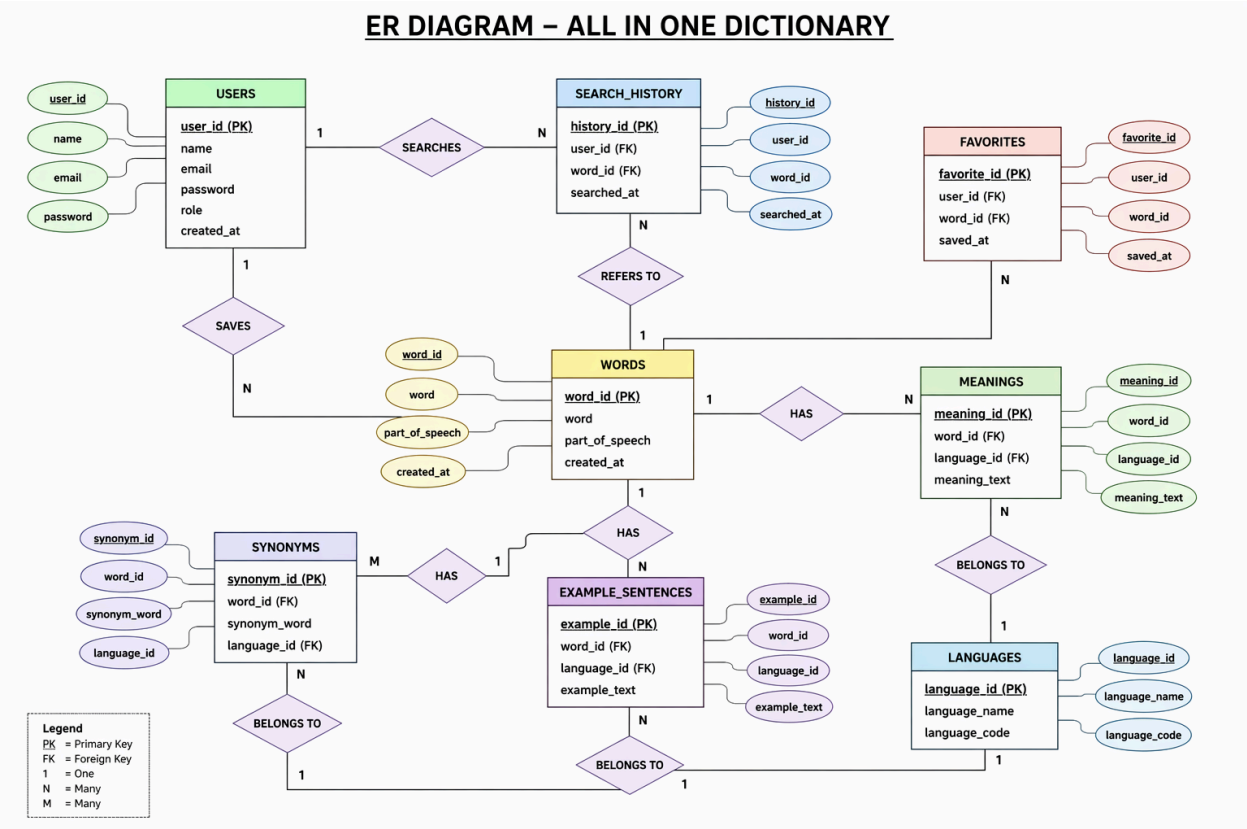
- Unicode language compatibility.
- Internationalization support.
- Large vocabulary database management.

6.1 Glossary

Term	Meaning
API	Application Programming Interface
TTS	Text-to-Speech
UI	User Interface
DB	Database

6.2 Analysis Models

❒Entity Relationship Diagram of A Multilingual Smart Dictionary System



To Be Determined

- Final supported languages list
- Offline storage limit
- Final API provider